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1. Find the vector with magnitude 39 in the same direction as (12,-5) [2 marks]

2. Given $a=(2,-1)$ and $b=(-4,5)$, find constants λ and μ such that $5a+\lambda(4,6)=\mu b$ [4 marks]

3. Given that the x-coordinate of B is less than the x-coordinate of D, write EB and ED as column vectors. [2 marks]

4. The parallelogram OABC has $OA = a$ and $OC = c$. Point D lies on CB such that $CD:DB = 3:1$. Lines AB and OD meet at E where $OE = hOD$ and $BE = kAB$. Find DE in terms of a, c and h. [4 marks]

5. Hence find the value of h and of k. [4 marks]

6. Triangle OAB, point P on AB with $AP:PB=1:3$. $OA=a$, $OB=b$. Find OP. [2 marks]

7. Vector $\mathbf{q}$: magnitude $12\sqrt{5}$, direction $(6, -3)$. Vector $\mathbf{r}$: magnitude $15\sqrt{2}$, direction $(-5, 5)$. Find unit vector in direction $\mathbf{q} + \mathbf{r}$. [6 marks]

8. $\mathbf{P}$ initially has position vector $(3, 5)$. Write down position vector of $\mathbf{P}$ at time t . [2 marks]

9. Find two expressions for $\mathbf{OP}$ and show that $\mathbf{P}$ divides both $\mathbf{AC}$ and $\mathbf{DB}$ in the ratio $4:7$. [7 marks]

10. The point $\mathbf{Q}$ is such that $\mathbf{OQ} = \frac{2}{7}\mathbf{b} + \frac{2}{7}\mathbf{c}$. Use a vector method to show that $\mathbf{O}$, $\mathbf{Q}$ and $\mathbf{P}$ are collinear. [2 marks]

Answer key

1. (36, -15)
2. $\lambda = -15/22$, $\mu = -20/11$
3. $EB = (-5, 3)$, $ED = (5, -3)$
4. $CD = 3a/4$, $OD = c + 3a/4$, $OE = h(c + 3a/4)$, $DE = h(c + 3a/4) - (c + 3a/4)$
5. Equate the two expressions for DE. $h - 1 = k$, $3h/4 = 1/4$. $h = 4/3$, $k = 1/3$
6. $OP = (3/4)a + (1/4)b$
7. $q = (24, -12)$, $r = (-15, 15)$, unit vector $= (1/\sqrt{90})(9, 3)$
8. position $= (3, 5) + t(-10, 10.5)$
9. $\lambda = 4/11$, $\mu = 7/11$
10. OP and OQ are scalar multiples